

An abstract network diagram with various sized nodes (black, blue, and grey) connected by thin grey lines. Some nodes are highlighted with larger circles. The background is white with faint grey circles.

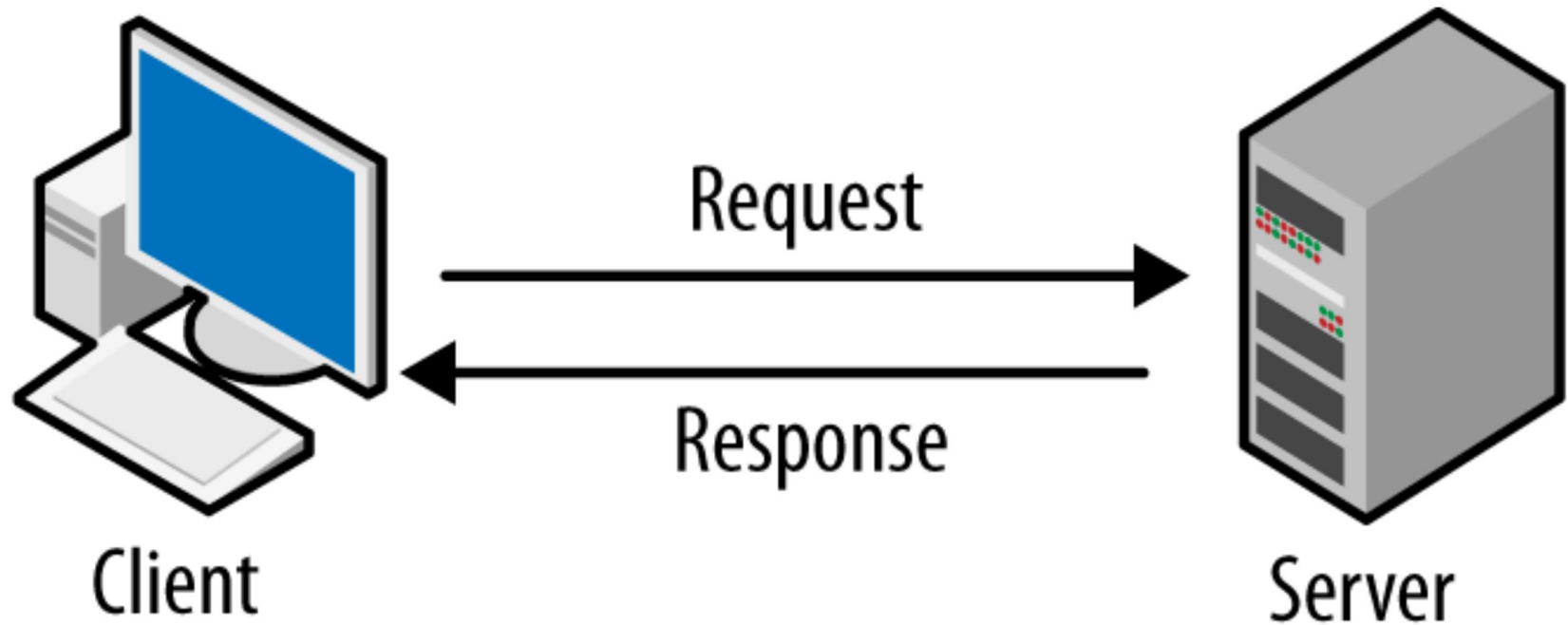
Fakultas Ilmu Komputer

Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

Fundamental Concept of Client-server Communications



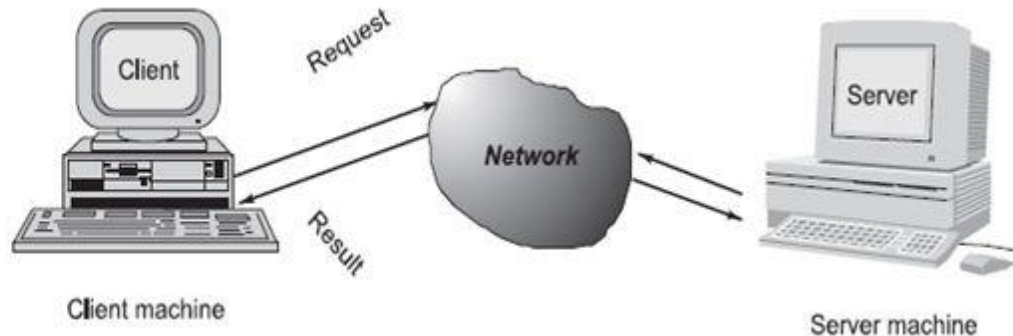
Client Server Communications



Client-server communication is a fundamental concept in computer networking and software architecture. Client-server communication shows the interaction between two types of entities: **clients** and **servers**. **Clients request** and consume resources or services provided by servers.

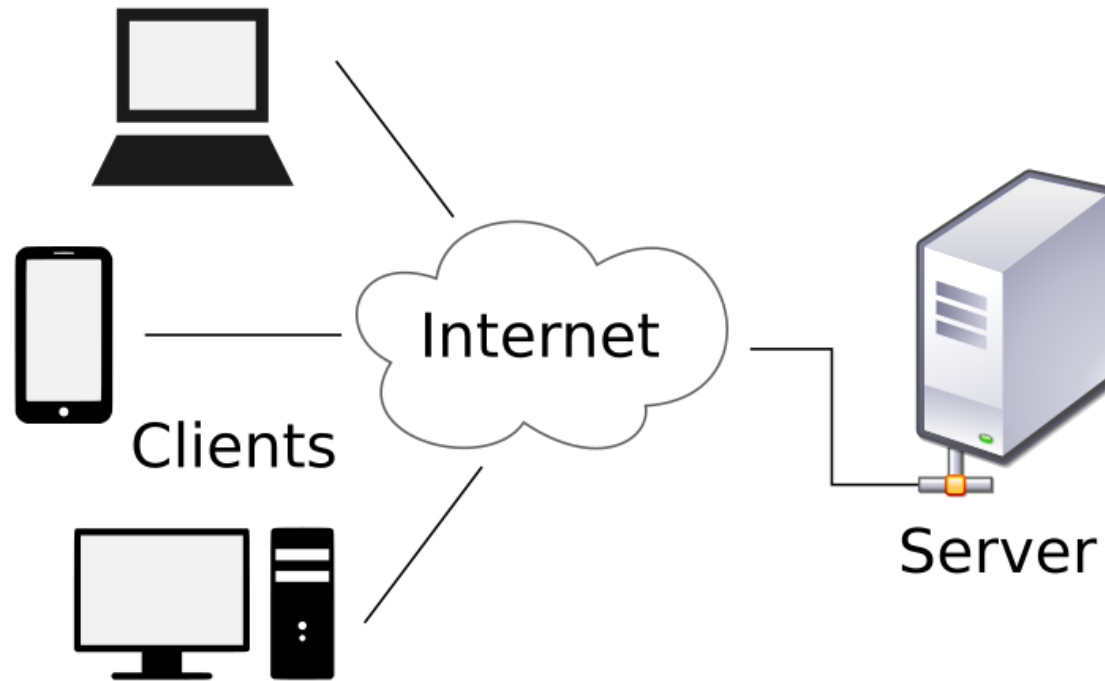
Client/Server Communication

- At a basic level, network-based systems consist of a server , client , and a media for communication as shown in Fig:



- A computer running a program that makes a request for services is called **client machine**.
- A computer running a program that offers requested services from one or more clients is called **server machine**.
- The media for communication can be wired or wireless network.

Client Server Communications



Servers are devices or software applications that provide services, data, or resources to clients. **Servers** are designed to (1) listen for incoming requests, (2) process those requests, and (3) send back appropriate responses/output.

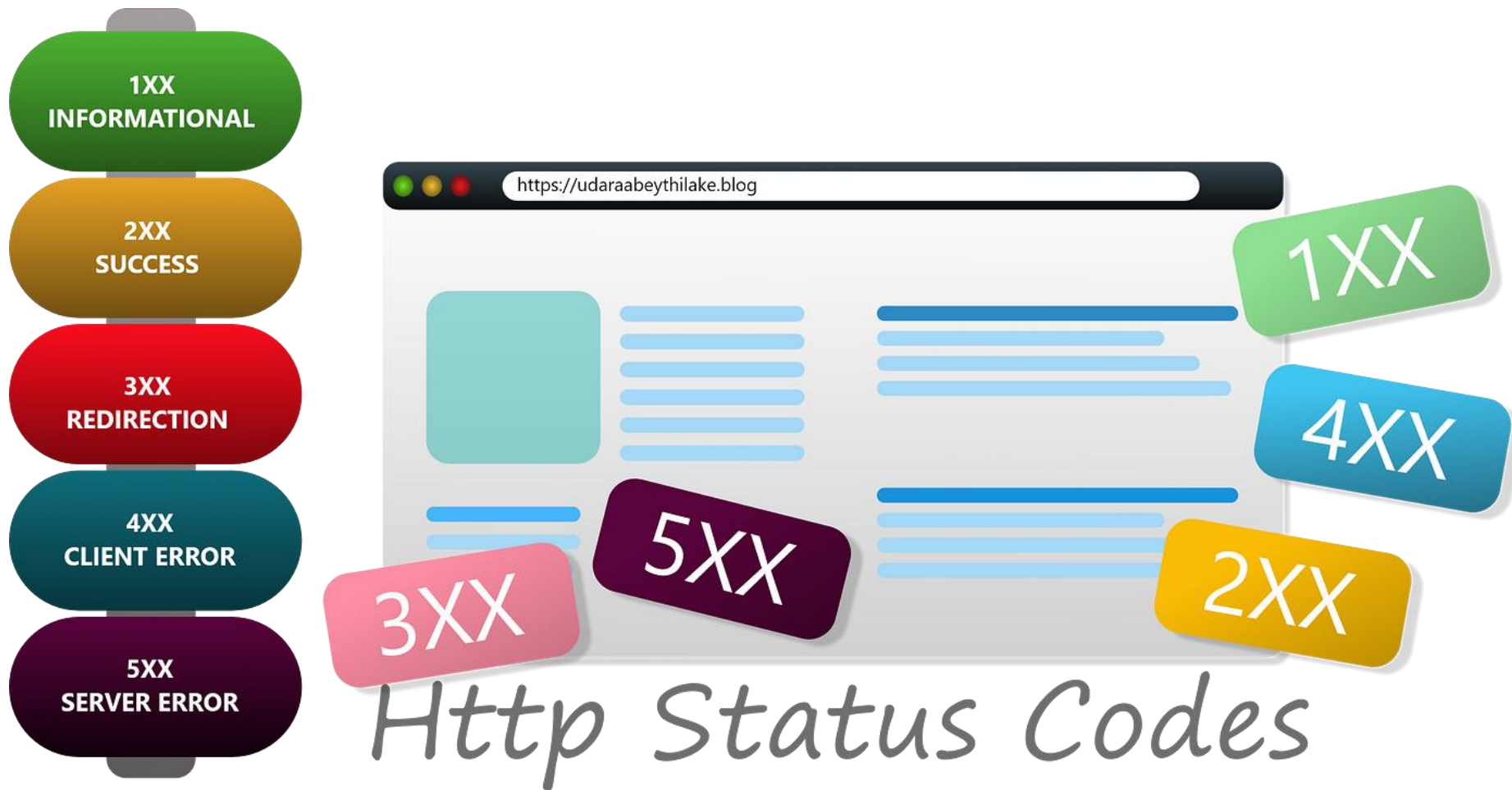
HTTP/HTTPS



Port 80 : Hypertext Transfer Protocol (**HTTP**)

Port 443 : Hypertext Transfer Protocol Secure (**HTTPS**)

HTTP & HTTPS are used for web browsing and communication between web browsers and web servers. **HTTP is the standard protocol**, while **HTTPS adds security through encryption**.



HTTP Status Codes are three-digit codes returned by the server in response to a request from a client (browser, API, or other application). These codes help in understanding whether the request was successful, encountered an error, or requires further action.

HTTP Status Codes

Successful



✓ GET, HEAD, PUT, POST

✓ 200 OK

201 Created (for POST)

204 No Content



Redirection



301 Moved Permanently

302 Found (temporarily)



Client Error



Incorrect Request

400 Bad Request

401 Unauthorized

403 Forbidden

404 Not Found



Server Error



500 Internal Server Error

501 Not Implemented

502 Bad Gateway

503 Service unavailable

504 Gateway Timeout



HTTP Status
Codes are three-
digit codes
returned by the
server

Elements Console Sources **Network** ⚠ Performance Memory Application Privacy and security

⏏ ⛔ 🔍 ☐ Preserve log ☐ Disable cache Fast 4G

100 ms 200 ms 300 ms 400 ms 500 ms 600 ms 700 ms

Name	Status
presensi.unklab.ac.id	200
aos.css	200
bootstrap.min.css	200
style.css	200

Name	Status	Type	Initiator
✖ hkhkhk	404	document	Other
📄 jquery-3.6.0.min.js	200	script	hkhkhk7

2 requests | 7.3 kB transferred | 96.2 kB resources | Finish: 1.71 s | DOMContentLoaded: 1.76 s | Load: 1.83 s

⋮ Console What's new AI assistance Issues

🔍 top 👁 Filter

✖ GET <https://presensi.unklab.ac.id/hkhkhk> 404 (Not Found)

> `ctrl` `i` to turn on code suggestions. Don't show again

Name	Status	Type	Initiator
✖ app	403	document	Other

1 requests | 5.6 kB transferred | 5.0 kB resources | Finish: 78 ms | DOMContentLoaded: 140 ms | Load: 149 ms

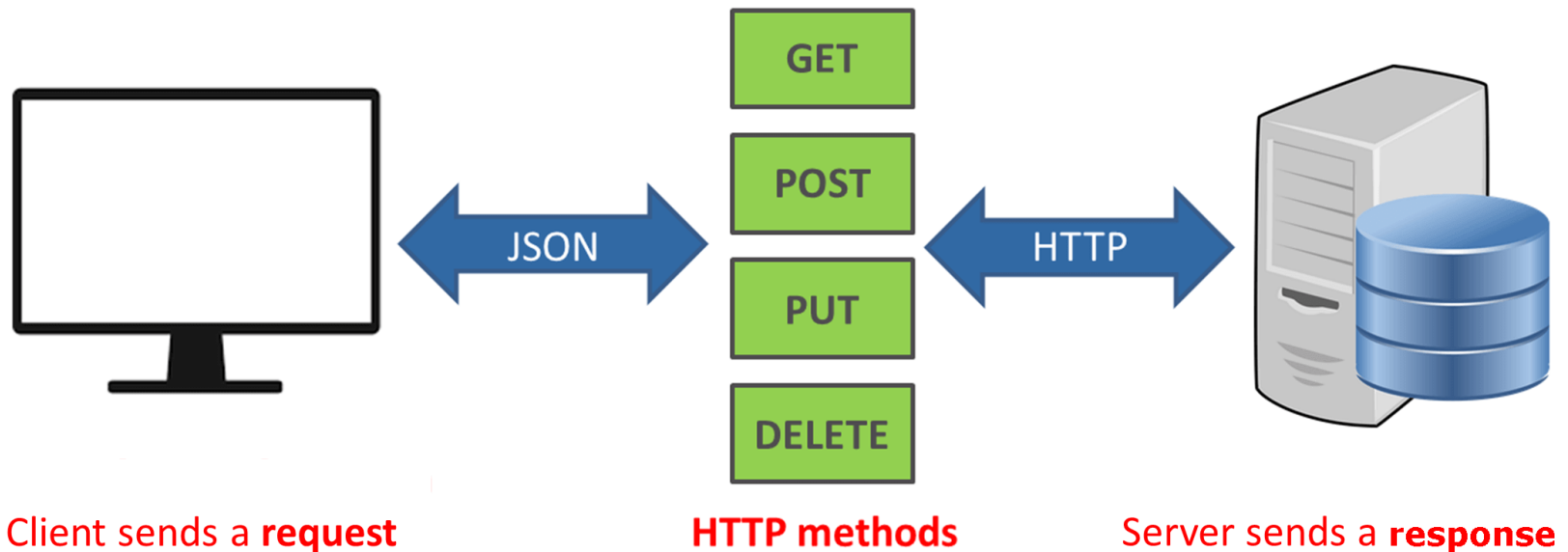
⋮ Console What's new AI assistance Issues

🔍 top 👁 Filter

✖ GET <https://presensi.unklab.ac.id/xyz/app> 403 (Forbidden)

> `ctrl` `i` to turn on code suggestions. Don't show again

REST & APIs



Representational State Transfer (REST) is an architectural style for designing networked applications. It involves using standard HTTP methods (GET, POST, PUT, DELETE) to interact with resources on servers. RESTful APIs provide a structured way for clients to communicate with servers, typically exchanging data in JSON or XML format.



Apakah Setiap Komunikasi Browser— Server Dicatat di Server ?

Server otomatis
dicatat (logged)?



TIDAK

Tidak semua komunikasi browser-server akan otomatis dicatat (logged).

Apache (Linux)

```
/var/log/apache2/access.log  
/var/log/apache2/error.log
```

Apa Saja Komunikasi Browser–Server?

Setiap request browser berisi:

- ✓ HTTP Method (GET, POST, PUT, dll)
- ✓ URL
- ✓ Headers
- ✓ Body (data form / JSON)
- ✓ IP client
- ✓ User-Agent ???
- ✓ Cookie

Apa yang Dicatat Server (Default)?

Web Server (Apache / Nginx), biasanya hanya:

- ✓ IP address
- ✓ Timestamp
- ✓ URL
- ✓ HTTP status
- ✓ User-Agent (**what is user-agent?**)

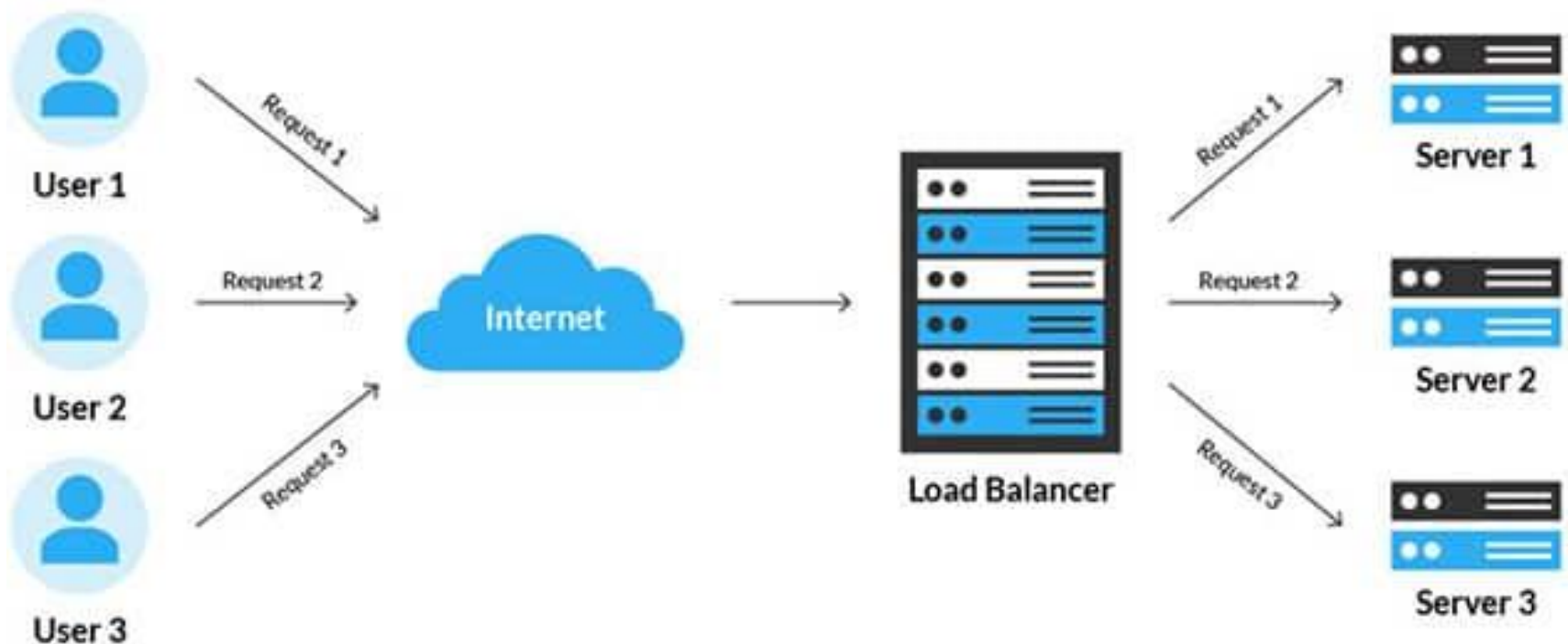
User-Agent HTTP header

User-Agent (**client identity description**) is an HTTP header that identifies the client's browser, operating system, and device to the server. It is used for compatibility, analytics, and **security purposes but it can be easily spoofed/dipalsukan**. The access.log is a web server log file that records incoming HTTP requests from browsers, apps, or bots.

```
192.168.1.4 - - [26/Apr/2018:17:18:54 +0200] "GET /2017/05/30/how-to-retrieve-iis-http-logs-remotely/ HTTP/1.1" 301 671 "-" "Mozilla/5.0 (Windows NT 6.1; WOW64; Trident/7.0; rv:11.0) like Gecko"
192.168.1.4 - - [26/Apr/2018:17:18:54 +0200] "GET /2017/05/30/how-to-retrieve-iis-http-logs-remotely/ HTTP/1.1" 200 21466 "-" "Mozilla/5.0 (Windows NT 6.1; WOW64; Trident/7.0; rv:11.0) like Gecko"
192.168.1.4 - - [26/Apr/2018:17:19:00 +0200] "GET /2017/05/ HTTP/1.1" 200 19215 "https://www.wptest.com/2017/05/30/how-to-retrieve-iis-http-logs-remotely/" "Mozilla/5.0 (Windows NT 6.1; WOW64; Trident/7.0; rv:11.0) like Gecko"
192.168.1.4 - - [26/Apr/2018:17:19:03 +0200] "GET /2017/05/30/hello-world/ HTTP/1.1" 200 19742 "https://www.wptest.com/2017/05/" "Mozilla/5.0 (Windows NT 6.1; WOW64; Trident/7.0; rv:11.0) like Gecko"
webmaster@USERVER:~$
```

What is Load Balancing?

Load balancing is a technique used to **distribute network traffic to multiple servers**. This distribution is done for no other reason than to lighten the workload so that **the server is not too heavy when there are many requests from clients**. With load balancing, the process of distributing network traffic to several servers will be more optimal, efficient, stable and fast.



What is Auto Scaling Server??



Amazon.com

<https://aws.amazon.com> › autoscaling

AWS Auto Scaling

AWS Auto Scaling monitors your applications and automatically adjusts capacity to maintain steady, predictable performance at the lowest possible cost.

[Amazon EC2 Auto Scaling](#) · [Features](#) · [Pricing](#) · [Resources](#)

People also ask

What is an Auto Scaling server?

Autoscaling is a cloud computing feature that enables organizations to scale cloud services such as server capacities or virtual machines up or down automatically, based on defined situations such as traffic or utilization levels.



The slide features a white background with green leaves and branches framing the top and bottom edges. The leaves are vibrant green with some showing signs of aging or damage. A large, black, serif-style title is centered on the slide.

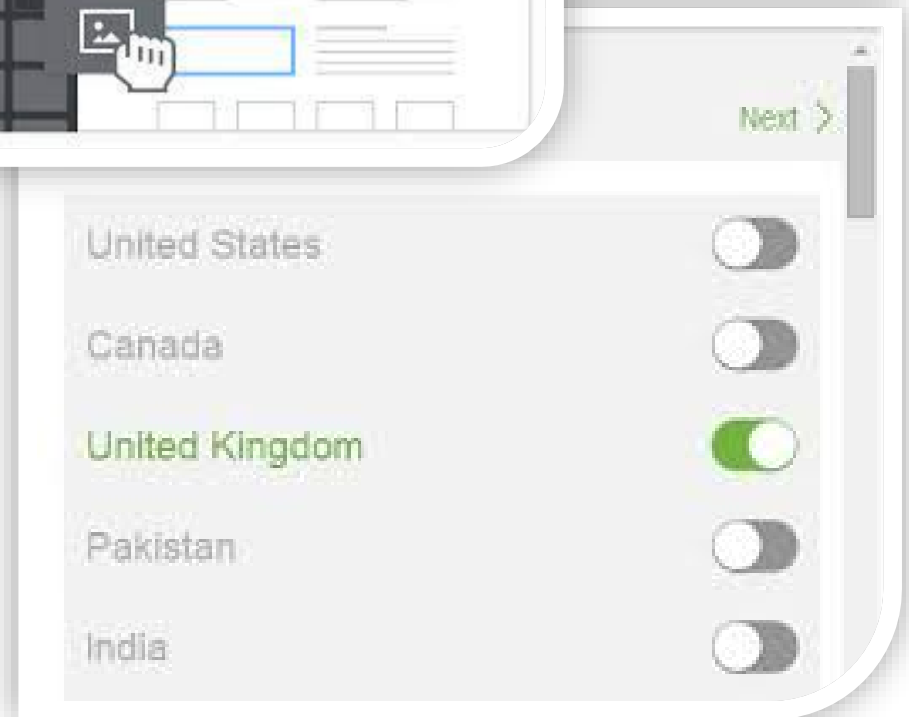
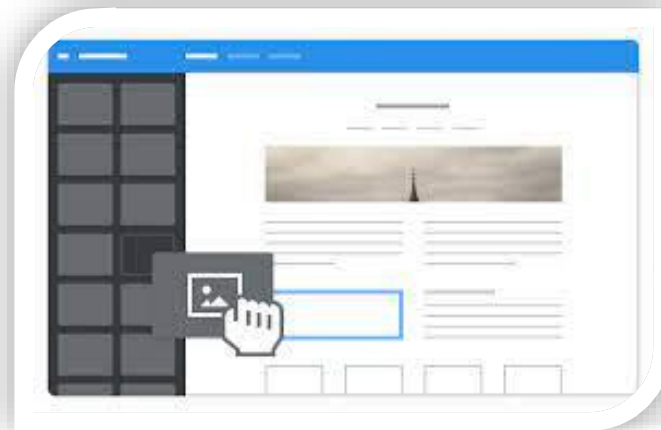
1 – jQuery Event Handling

What *are* Events?

- ❖ All the different visitors' actions that a web page can respond to are called events.
- ❖ An event represents the precise moment when something happens.

Examples:

- 1) A mouse click.
- 2) A web page loading.
- 3) Taking mouse over an element.
- 4) Submitting an HTML form.
- 5) A keystroke on your keyboard,
- 6) etc.

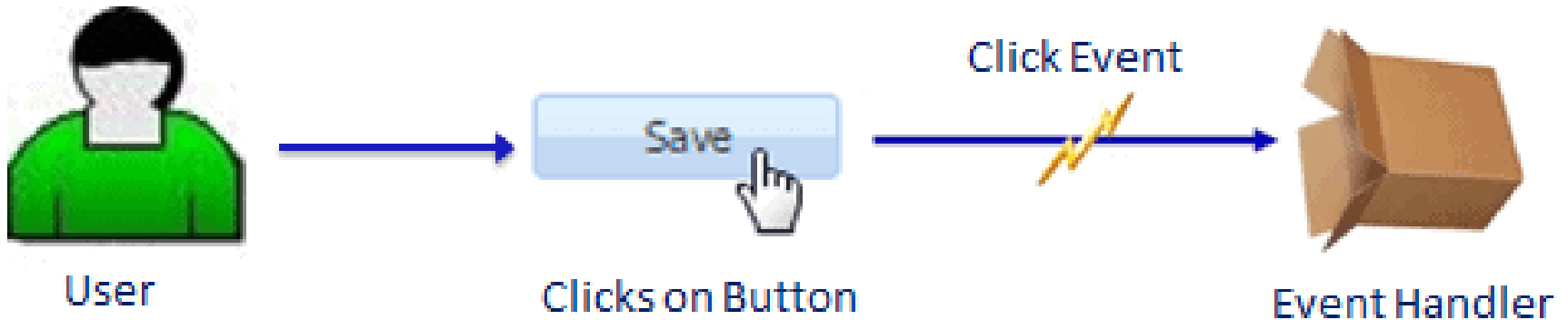


What are jQuery Events?



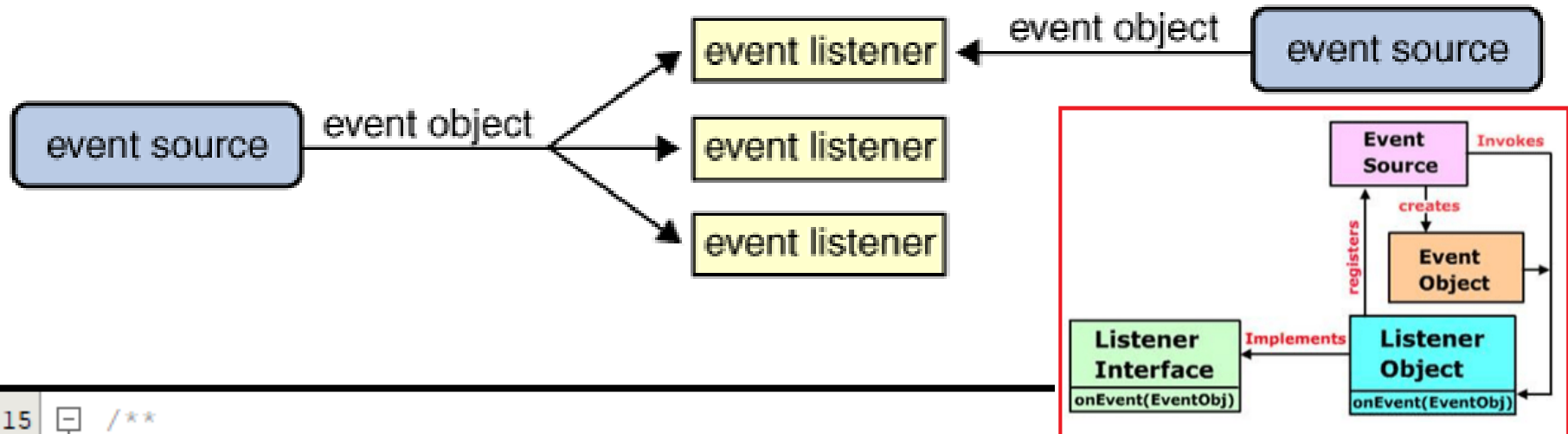
- 1) A jQuery Event is the result of **an action that can be detected by jQuery** (JavaScript).
- 2) When these events are triggered, we can then use a **custom function to do pretty much whatever we want with the event.**
- 3) These custom functions are called **Event Handlers.**
- 4) The jQuery library **provides methods to handle all the DOM events** and make complete event handling considerably easier than what we have available in JavaScript.

jQuery Events



In most web applications, the *user* does some action to perform an operation. For example, *user* clicks on save button to save the edited data in a web page. When, clicking on the button is a user's action, which **triggers click event** and click event handler (*function*) saves data.

Java Action Event Handling



```
15  /**
16   ClickButtonEvent is not abstract and does not override abstract method actionPerformed(ActionEvent) in A
17   ----
18   (Alt-Enter shows hints)
19
20  public class ClickButtonEvent implements ActionListener {
21      Implement all abstract methods
22      Make class ClickButtonEvent abstract
23      static ClickButtonEvent listener = new ClickButtonEvent();
24      static JLabel label;
25
26      /**
27       * @param args the command line arguments
28       */
29      public static void main(String[] args) {
30          JFrame win = new JFrame("Button Events").
```

Java Action Event Handling

```
1 import java.awt.event.*; // package for user events
2 import javax.swing.*;
3 import java.awt.*;
4
5 public class ChangeTitle extends JFrame implements ActionListener {
6     JButton b1 = new JButton("Rosencrantz");
7     JButton b2 = new JButton("Guildenstern");
8
9     public ChangeTitle() {
10         super("Title Bar");
11
12         b1.addActionListener(this); //listener for Rosencrantz button
13         b2.addActionListener(this); //listener for Guildenstern button
14         JPanel pane = new JPanel();
15         pane.add(b1);
16         pane.add(b2);
17
18         setContentPane(pane);
19     }
20
21     public static void main(String[] arguments) {
22         JFrame frame = new ChangeTitle();
23
24         ExitWindow exit = new ExitWindow();
25         frame.addWindowListener(exit);
26
27         frame.pack();
28         frame.setVisible(true);
29     }
30
31     public void actionPerformed(ActionEvent evt) {
32         Object source = evt.getSource(); //determine user action
33         if (source == b1)
34             setTitle("Rosencrantz");
35         else if (source == b2)
36             setTitle("Guildenstern");
37         repaint(); // redraw frame to display change
38     }
39 }
40
41 class ExitWindow extends WindowAdapter {
42     public void windowClosing(WindowEvent e) {
43         System.exit(0);
44     }
45 }
```

ActionListener in Java is a class that is responsible for handling all action events such as when the user clicks on a component. Mostly, action listeners are used for JButtons.

An ActionListener can be used by the implements keyword to the class definition. It can also be used separately from the class by creating a new class that implements it. It should also be imported to your project.

Common DOM Events *in* JQuery

Mouse Events	Keyboard Events	Form Events	Document/Window Events
click	keypress	submit	load
dblclick	keydown	change	resize
mouseenter	keyup	focus	scroll
mouseleave		blur	unload

Mouse Events	Keyboard Events	Form Events	Document Events
click	keypress	submit	load
dblclick	keydown	change	resize
hover	keyup	select	scroll
mousedown		blur	unload
mouseup		focusin	ready

Common DOM Events *in* JQuery

Mouse Events	Keyboard Events	Form Events	Document Events
click	keypress	submit	load
dblclick	keydown	change	resize
hover	keyup	select	scroll
mousedown		blur	unload
mouseup		focusin	ready



- 1) The **click() method** attaches an event handler function to an HTML element. The function is executed when the user clicks on the HTML element.
- 2) The **dblclick() method** attaches an event handler function to an HTML element. The function is executed when the user double-clicks on the HTML element.
- 3) The **mouseenter() method** attaches an event handler function to an HTML element. The function is executed when the mouse pointer enters the HTML element.
- 4) The **mouseleave() method** attaches an event handler function to an HTML element. The function is executed when the mouse pointer leaves the HTML element.
- 5) The **mousedown() method** attaches an event handler function to an HTML element. The function is executed, when the left, middle or right mouse button is pressed down, while the mouse is over the HTML element.
- 6) The **hover() method** takes two functions and is a combination of the `mouseenter()` and `mouseleave()` methods.
- 7) The **on() method** attaches one or more event handlers for the selected elements.

jQuery Example (Handle Button Click Event)

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <script src="https://ajax.googleapis.com/ajax/libs/jquery/1.11.2/jquery.min.js">
5   </script>
6   <script>
7     $(document).ready(function () {
8
9       $('#btn-delete').click(function () {
10         alert('Delete button clicked !!!');
11       });
12
13     });
14   </script>
15 </head>
16 <body>
17   <h1>Demo: jQuery click() method</h1>
18
19   <input type="button" value="Delete Button" id="btn-delete" />
20 </body>
21 </html>
```

To handle DOM events using jQuery methods, first get the reference of DOM element(s) using jQuery selector and invoke appropriate jQuery event method.

jQuery mouseenter() & mouseleave() method

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <script src="https://ajax.googleapis.com/ajax/libs/jquery/1.11.2/jquery.min.js">
5   </script>
6   <script>
7     $(document).ready(function () {
8       // handle mouse enter
9       $('#myDiv').mouseenter(function (data) {
10         $(this).css('background-color', 'green');
11       });
12       // handle mouse leave
13       $('#myDiv').mouseleave(function (data) {
14         $(this).css('background-color', 'red');
15       });
16     });
17   </script>
18 </head>
19 <body>
20   <h1>Demo: jQuery mouseenter() & mouseleave() method </h1>
21   <div id="myDiv" style="width:100px;height:100px;background-color:red;">
22   </div>
23 </body>
24 </html>
```

jQuery Event binding

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <script src="https://ajax.googleapis.com/ajax/libs/jquery/1.11.2/jquery.min.js">
5   </script>
6   <script>
7     $(document).ready(function () {
8       // handle event binding
9       $('#myDiv').on('click', ':button', function () {
10         alert('Button clicked');
11       });
12     });
13   </script>
14 </head>
15 <body>
16   <h1> Demo: Event binding </h1>
17   <div id="myDiv" >
18     <input type="button" value="Save" id="saveBtn" />
19
20     <input type="button" value="Add" id="addBtn" />
21   </div>
22
23   <input type="button" value="Delete" id="delBtn" />
24 </body>
25 </html>
```

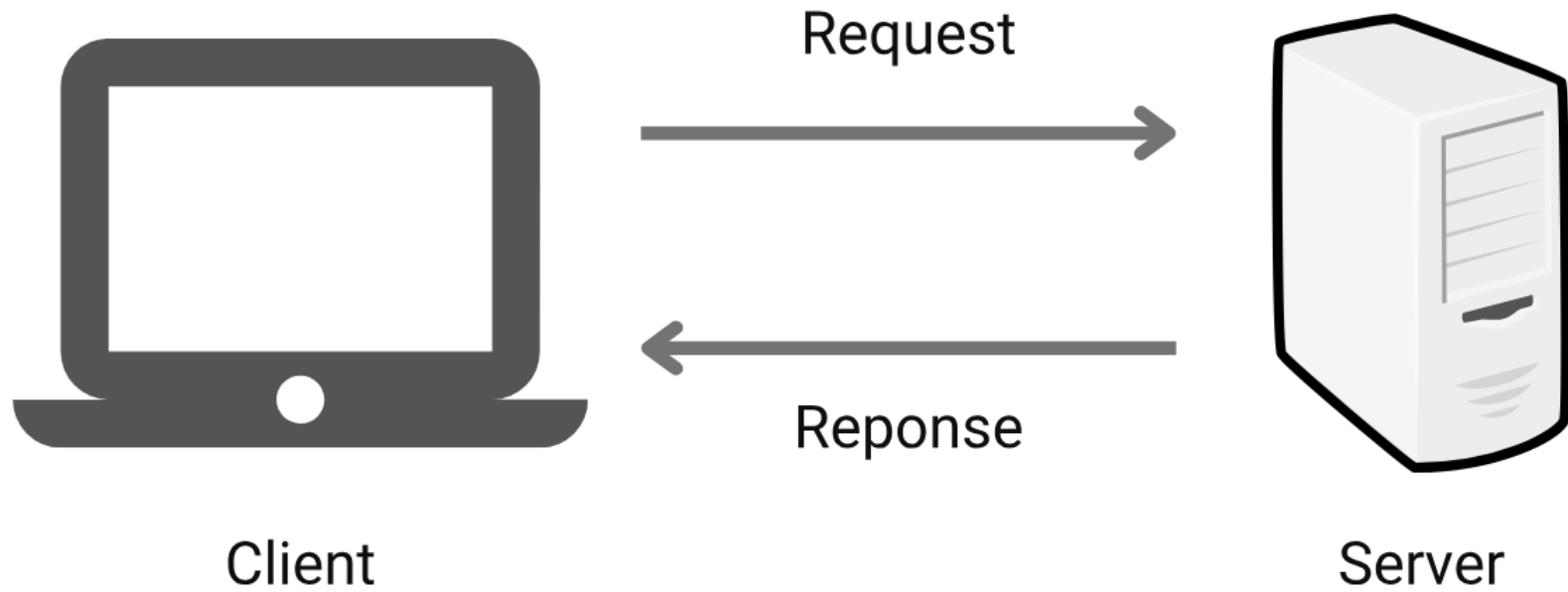
We specify **':button' selector**. So click event triggered by buttons in `<div>` tag whose *id* is **myDiv**, will only be handled.

A decorative border of green leaves and branches at the top of the slide.

2 – jQuery Ajax

A decorative border of green leaves and branches at the bottom of the slide.

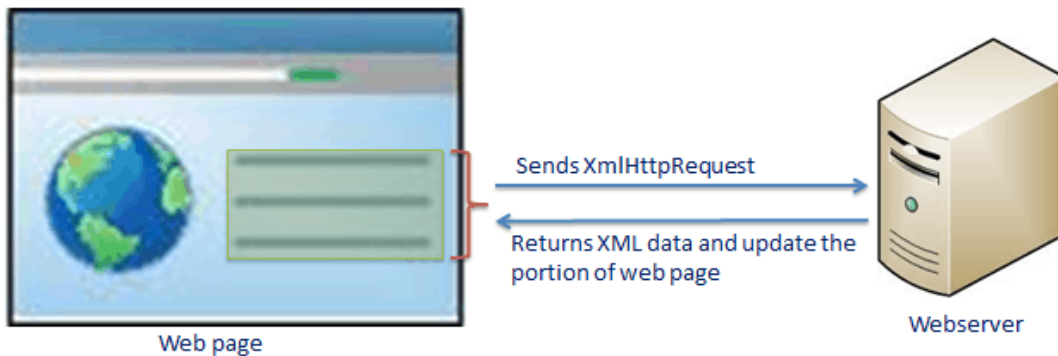
Client Server Communication



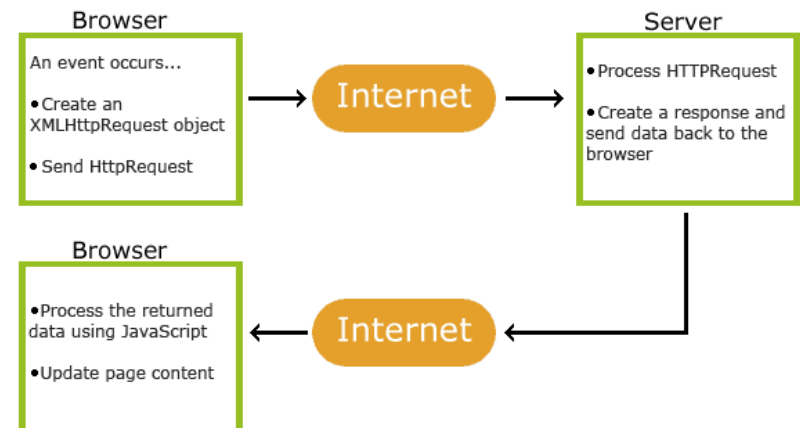
“**Client Server Communication**” memungkinkan *aplikasi* atau *user device* untuk berinteraksi dengan *server* yang menyediakan *service*, seperti mendapatkan data dari *server*, mengirim data ke *server*, atau menjalankan operasi yang membutuhkan *resource* dari *server*. Interaksi client-server ini memungkinkan aplikasi untuk memanfaatkan fungsionalitas yang disediakan oleh *server*, seperti pengolahan data, *storage*, atau *business logic* (*logika bisnis*).

What is AJAX?

- ❖ AJAX = Asynchronous (*asiNGkrənəs*) JavaScript and XML.
- ❖ In short; AJAX is about **loading data in the background** and display it on the webpage, **without reloading the whole page**.
- ❖ Examples of applications using AJAX: Gmail, Google Maps, Youtube, and Facebook tabs.



Dalam konteks AJAX, XML (eXtensible Markup Language) digunakan sebagai format data yang umum digunakan untuk pertukaran data antara server dan klien. XML adalah bahasa markup yang dirancang untuk menyimpan dan mengangkut data secara struktural.



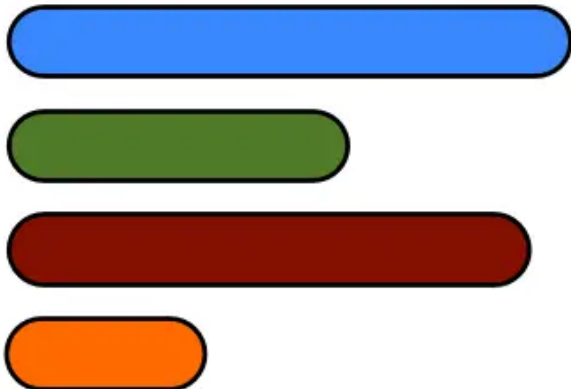
Meaning of Term "Asynchronous"

Asynchronous (asiNGkrənəs) JavaScript adalah kemampuan JavaScript untuk melakukan tugas-tugas secara “**non-blocking**” atau **tanpa harus menunggu** tugas yang sedang berjalan selesai sebelum melanjutkan ke tugas berikutnya. Dalam konteks pemrograman, JavaScript dapat **menjalankan beberapa tugas secara bersamaan** atau mengatur tugas-tugas untuk dijalankan di *background* sambil melanjutkan eksekusi kode lainnya.

Synchronous



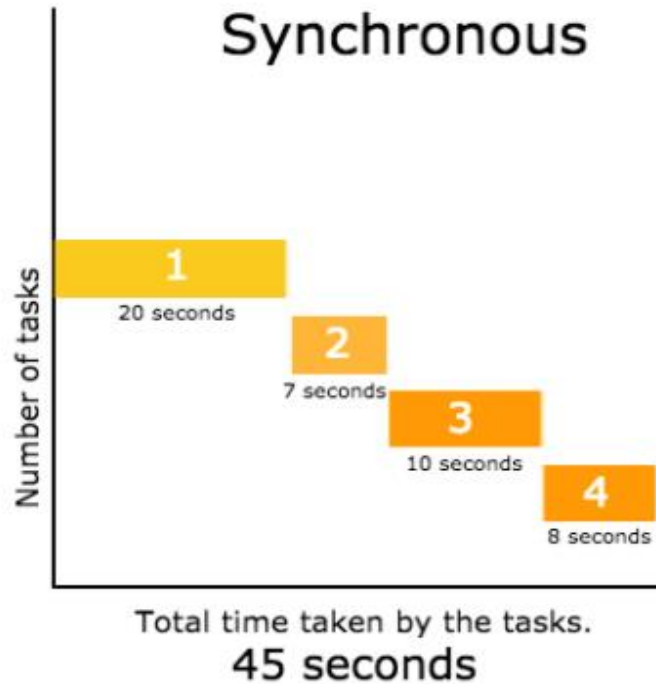
Asynchronous



Perbedaan utama antara asynchronous dan synchronous adalah dalam cara JavaScript mengelola tugas-tugas. Proses mana yang membutuhkan banyak waktu?



Synchronous vs Asynchronous JavaScript



Dalam mode *synchronous* (*sinkron*), JavaScript mengeksekusi tugas satu per satu **secara berurutan** dan mungkin harus menunggu tugas sebelumnya selesai sebelum melanjutkan ke tugas berikutnya.



Dalam mode *asynchronous* (*asinkron*), JavaScript dapat menjalankan tugas-tugas **secara paralel** atau menjalankan tugas-tugas tertentu di latar belakang tanpa harus menunggu tugas sebelumnya selesai. Ini dicapai dengan menggunakan konsep **callback functions**, **promises**, atau **async/await**.

How to handle “RESPONSE” from server?

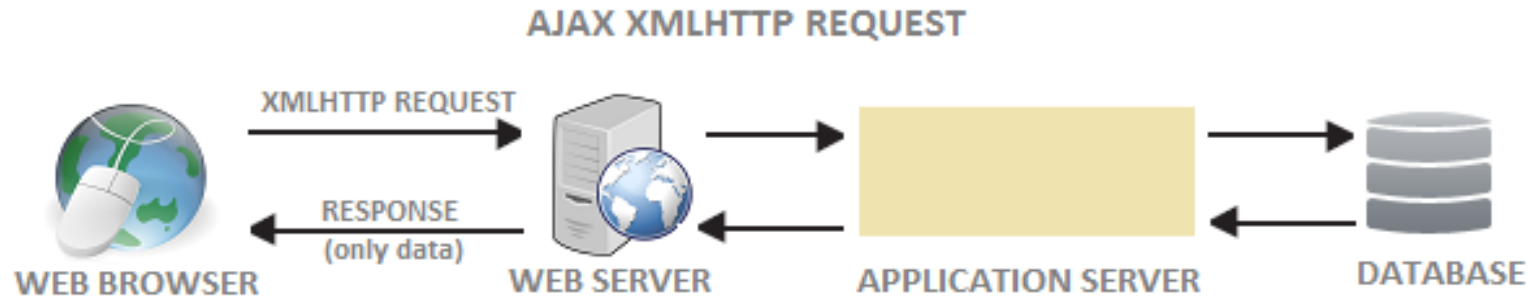
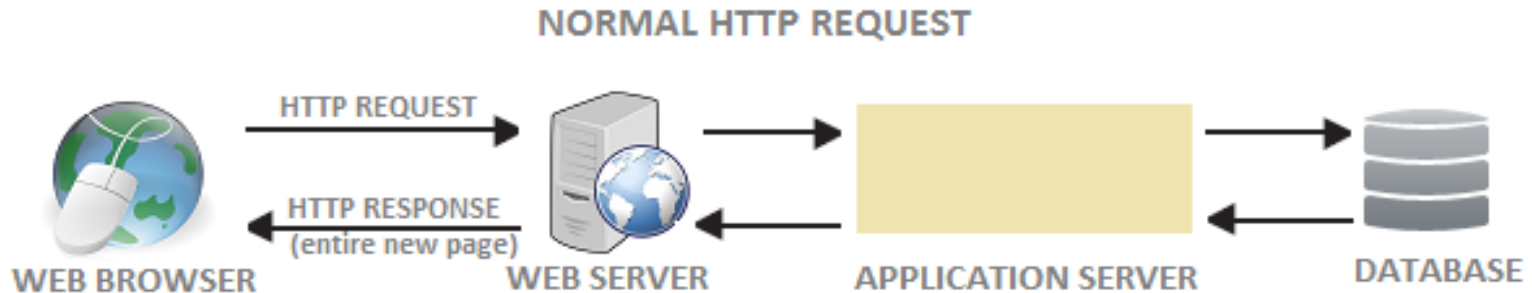
Ketika menggunakan AJAX (Asynchronous JavaScript and XML) untuk berkomunikasi dengan *server* atau sumber data eksternal, terdapat tiga pendekatan yang umum digunakan untuk mengelola tugas *asynchronous/asinkronas* (*menangani respons dari server*) :

Callback functions adalah fungsi yang diteruskan sebagai argumen kepada fungsi lain. Dalam konteks AJAX, callback functions digunakan untuk menangani respons atau hasil dari permintaan yang dilakukan ke *server*.

Promises adalah konstruksi bahasa JavaScript yang digunakan untuk mengelola tugas-tugas asinkronus dan menghasilkan nilai di masa depan. Dalam konteks AJAX, promises digunakan untuk mengelola permintaan dan menangani respons yang diterima dari *server*.

Async/await adalah fitur yang diperkenalkan dalam ES2017 (*atau ECMAScript 2017 adalah standar scripting language yang digunakan oleh bahasa pemrograman JavaScript*) yang menyederhanakan penulisan dan pengelolaan tugas asinkronus menggunakan promises. Dalam konteks AJAX, async/await digunakan untuk menulis kode asinkronus dalam gaya yang lebih sinkron dan mudah dibaca.

What *about* jQuery & AJAX?



Dengan menggunakan pendekatan *asynchronous*, JavaScript dapat **meningkatkan responsivitas aplikasi/web** dan menghindari terjadinya kondisi yang disebut "*blocking*", di mana ***eksekusi kode terhenti*** dan menunggu tugas selesai. Ini terutama penting ketika melakukan operasi yang membutuhkan waktu, seperti mengambil data dari *server*, mengirim permintaan HTTP, atau membaca dan menulis *file*.

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https://unkpresent.site/unklab/public/students/attendance/17143

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Class Attendance Checklist

Course: [IFMI352] Back-End Web Development - A

New + Summary Back →

Jadwal *attendance* yang di share secara PRIVATE tidak termasuk dalam hitungan TOTAL PERTEMUAN.

	Course Topic	Time	Radius	QR	Lock	Actions
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Tebaklah, kira-kira proses mana yang membutuhkan implementasi dari AJAX?

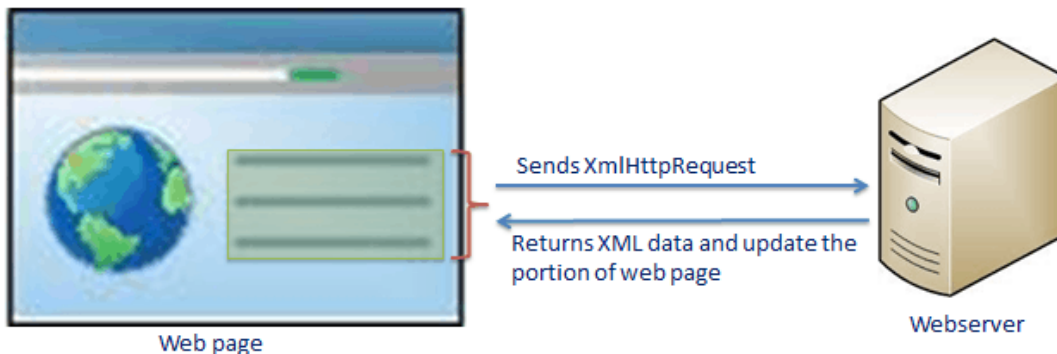
What *about* jQuery & AJAX?



- 1) jQuery provides **several methods for AJAX functionality**. With the jQuery AJAX methods, we can request text, HTML, XML, or JSON from a remote server using both HTTP Get and HTTP Post, and we can load the external data directly into the selected HTML elements of your web page.
- 2) Without jQuery, AJAX coding can be a bit tricky. Writing regular AJAX code can be a bit tricky, because different browsers have different syntax for AJAX implementation. This means that we will have to write extra code to test for different browsers.
- 3) The jQuery team has taken care of this, so we can write AJAX functionality with only one single line of code.

jQuery AJAX Methods

jQuery Methods	Ajax	Description
ajax()		Sends asynchronous http request to the server.
get()		Sends http GET request to load the data from the server.
Post()		Sends http POST request to submit or load the data to the server.
getJSON()		Sends http GET request to load JSON encoded data from the server.
getScript()		Sends http GET request to load the JavaScript file from the server and then executes it.
load()		Sends http request to load the html or text content from the server and add them to DOM element(s).



The jQuery library includes various methods to send Ajax requests. These methods internally use XMLHttpRequest object of JavaScript.

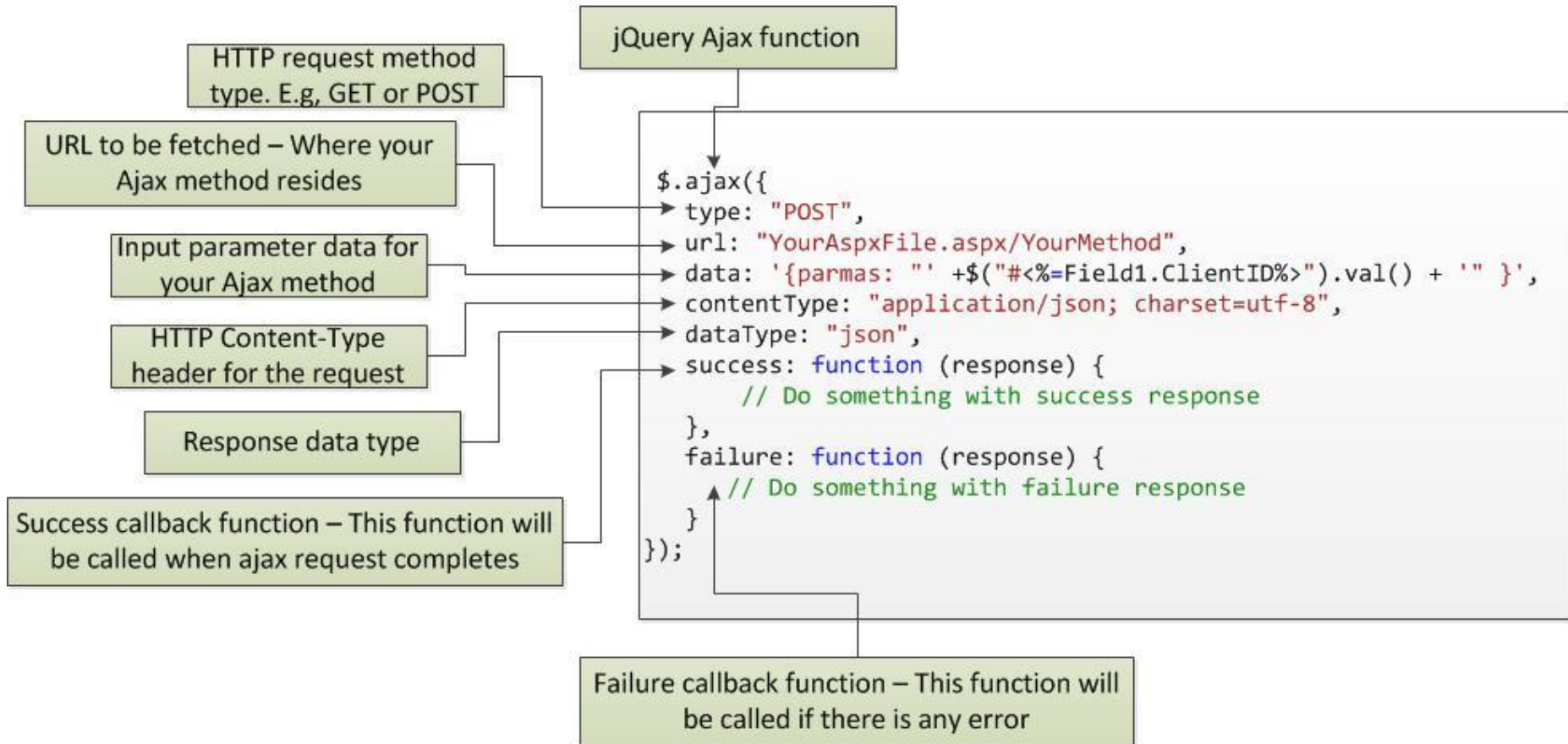
Example jQuery AJAX Methods

The diagram illustrates the mapping of jQuery AJAX method options to their respective labels. Red arrows point from the labels to the corresponding values in the code snippet:

- Request Type** points to `type: "POST",`
- Page Url** points to `url: "Default.aspx/MethodName",`
- Method Name** points to `MethodName` within the URL.
- Value** points to `"value"` within the data object.
- Parameter** points to `{param: "value" }` within the data object.
- Success Method** points to `success: OnSuccess,`

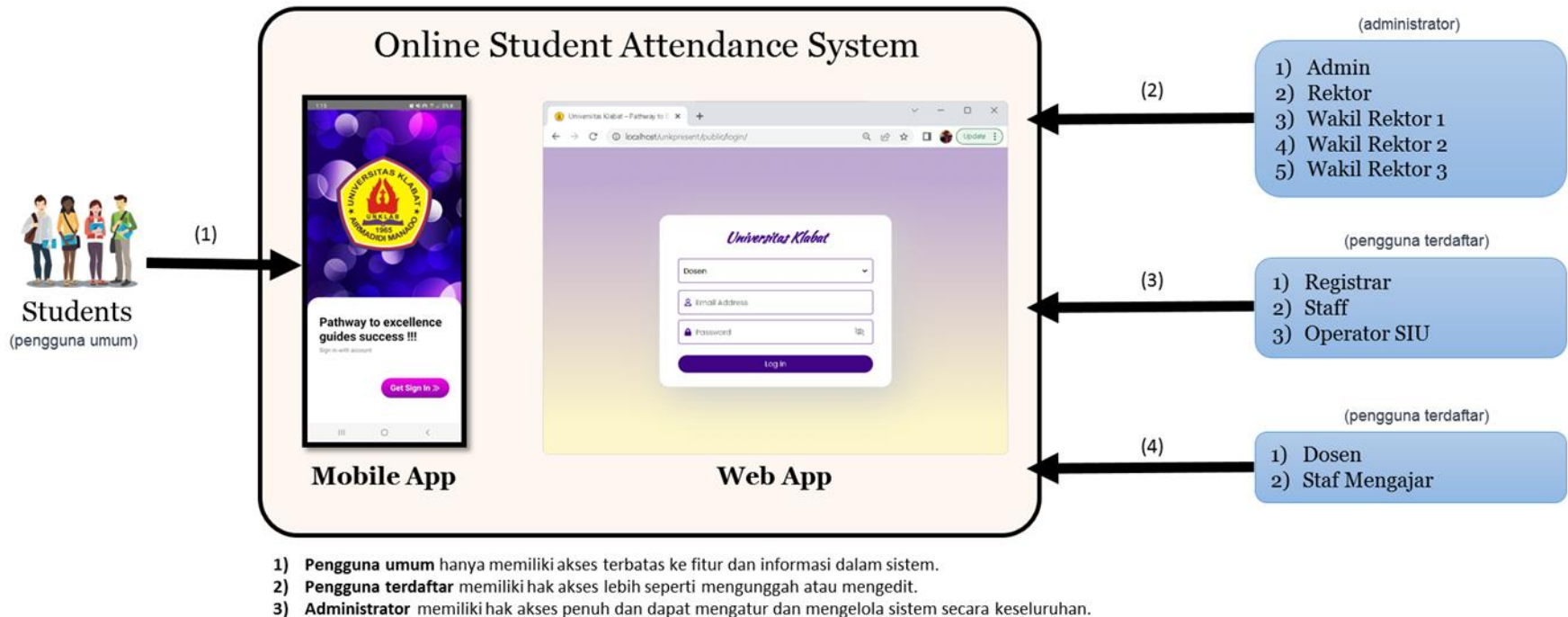
```
$.ajax({  
  type: "POST",  
  url: "Default.aspx/MethodName",  
  data: '{param: "value" }',  
  contentType: "application/json; charset=utf-8",  
  dataType: "json",  
  success: OnSuccess,  
});
```

Example jQuery AJAX Methods



Advantages of jQuery Ajax

- 1) Cross-browser support.**
- 2) Simple methods to use.**
- 3) Ability to send GET and POST requests.**
- 4) Ability to Load JSON, XML, HTML or Scripts.**



Contoh diatas merupakan pengembangan sistem presensi dua platform (*mobile* dan *web*), menggunakan *react native* dan PHP framework. Bagaimana cara **mobile app** mengirim dan/atau menerima data dari *server side*??? Apakah menggunakan AJAX??



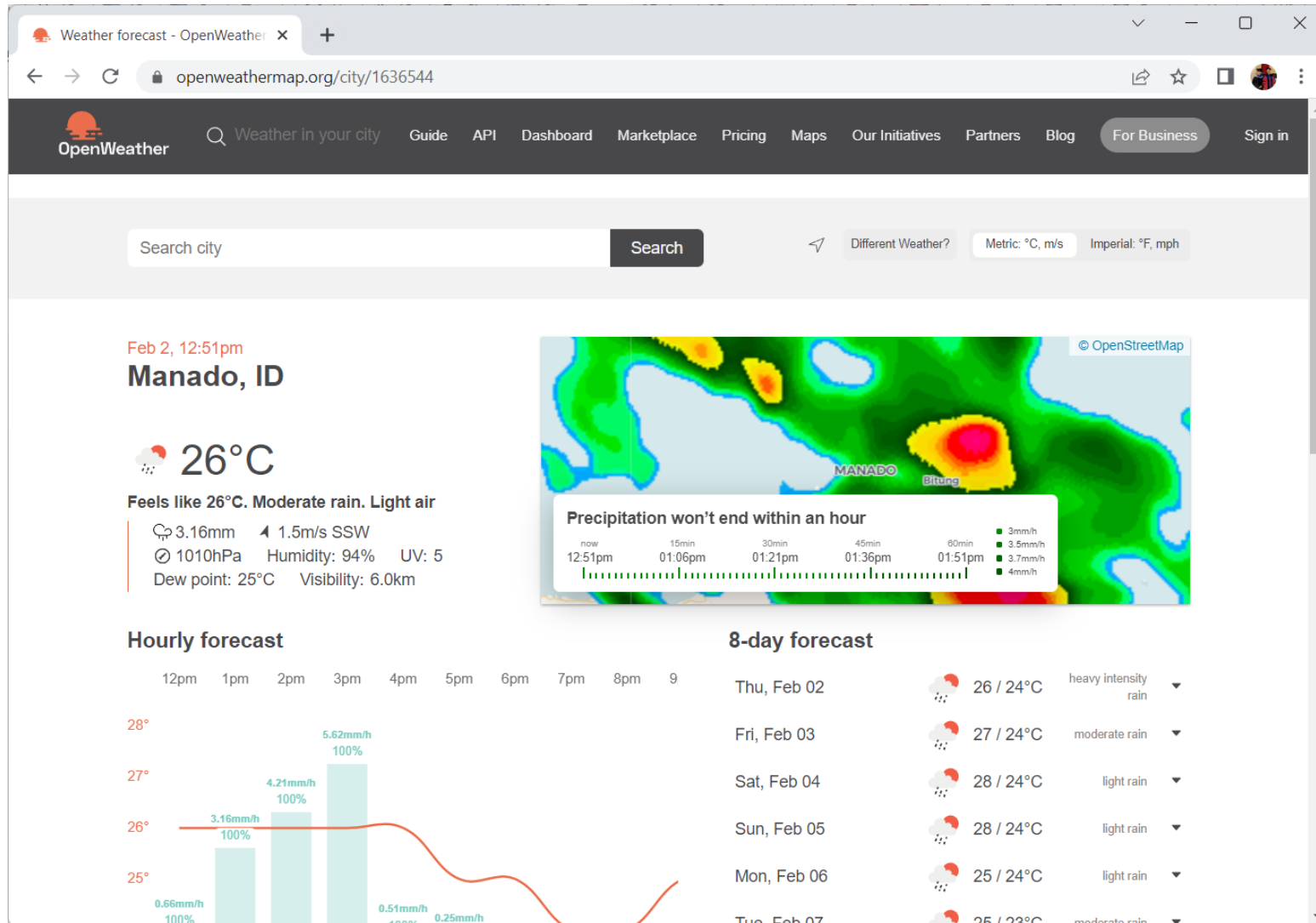
Apakah ada masalah keamanan jika kita menggunakan jQuery dan AJAX? Are there any security issues if we use jQuery and AJAX?



Exercise #1 *for* Students

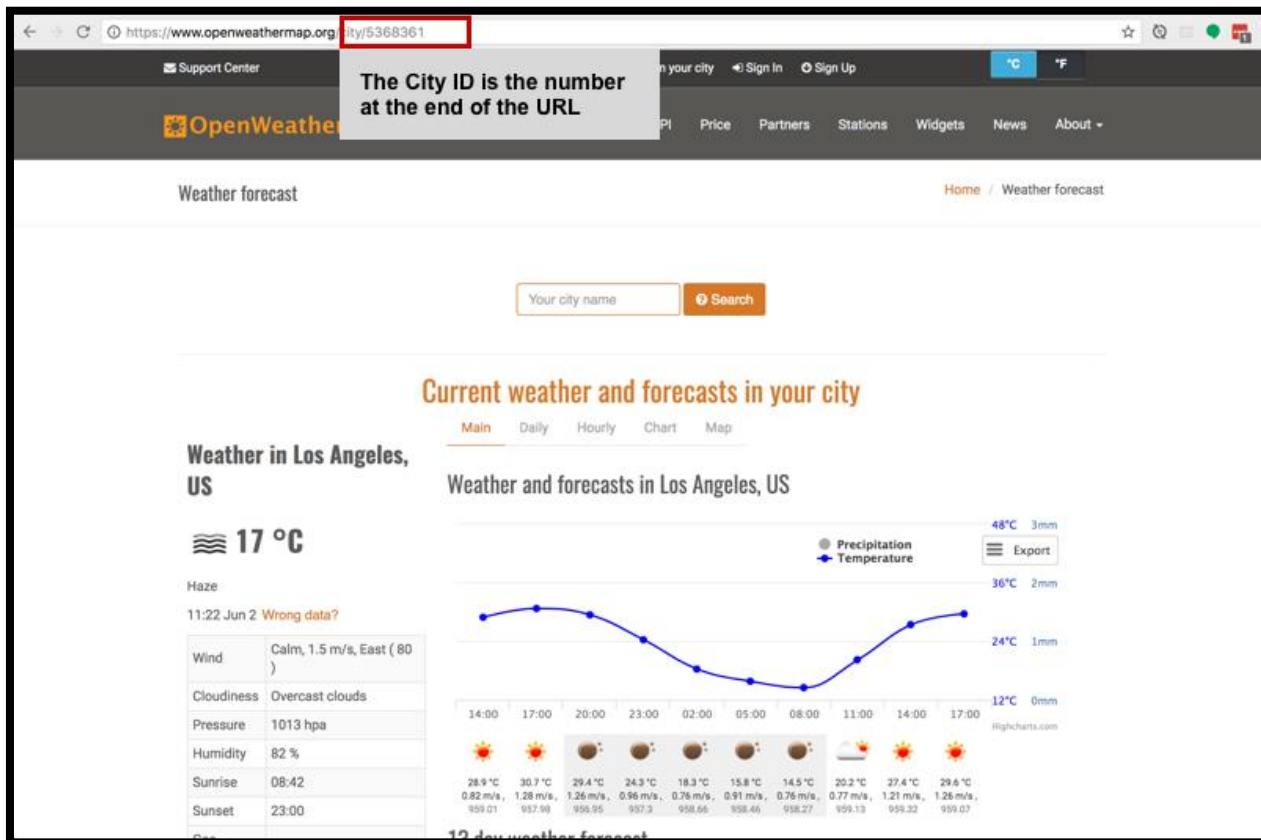
OpenWeatherMap API

Link website: <https://openweathermap.org/>

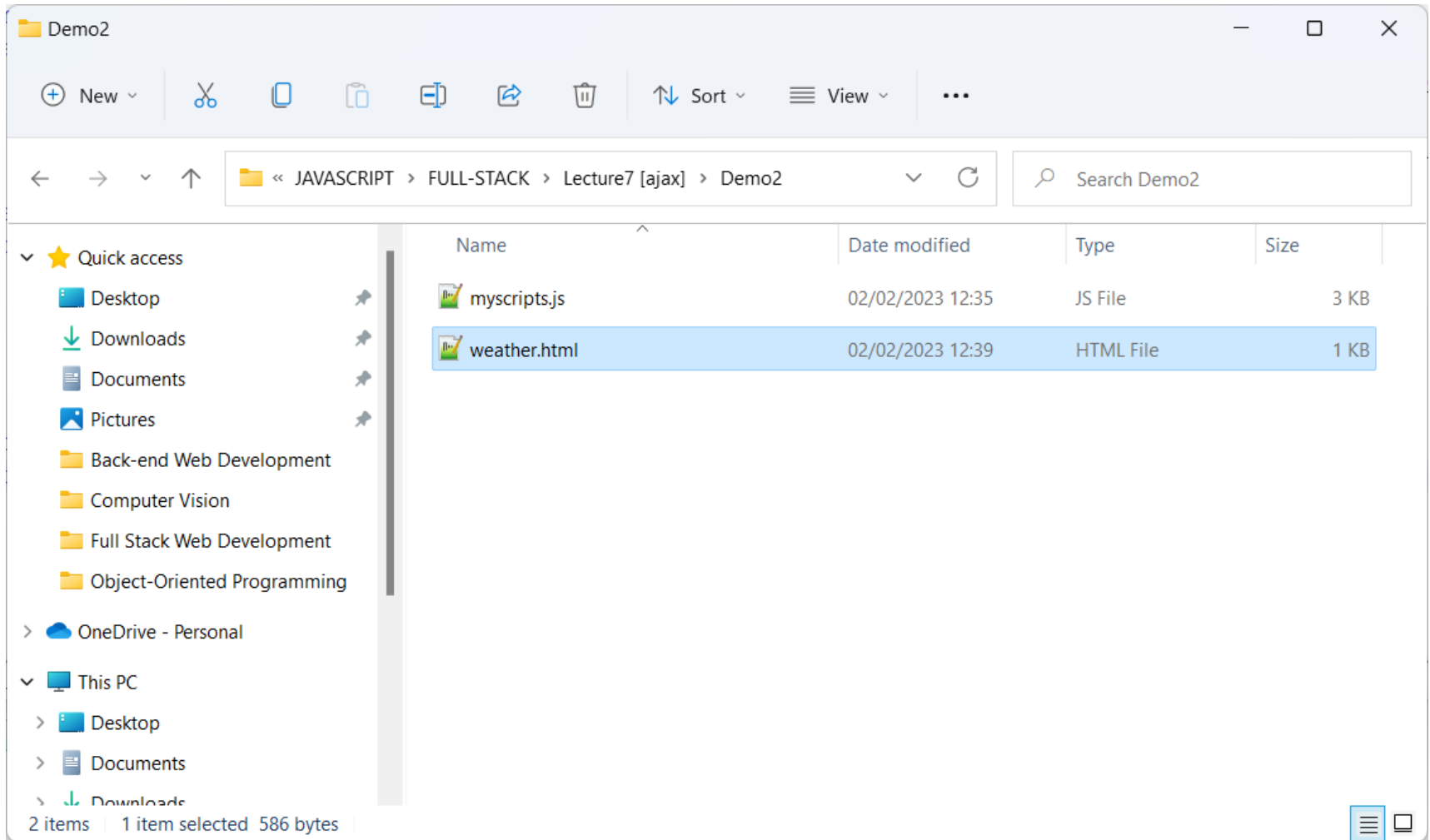


Get City ID *from* OpenWeatherMap

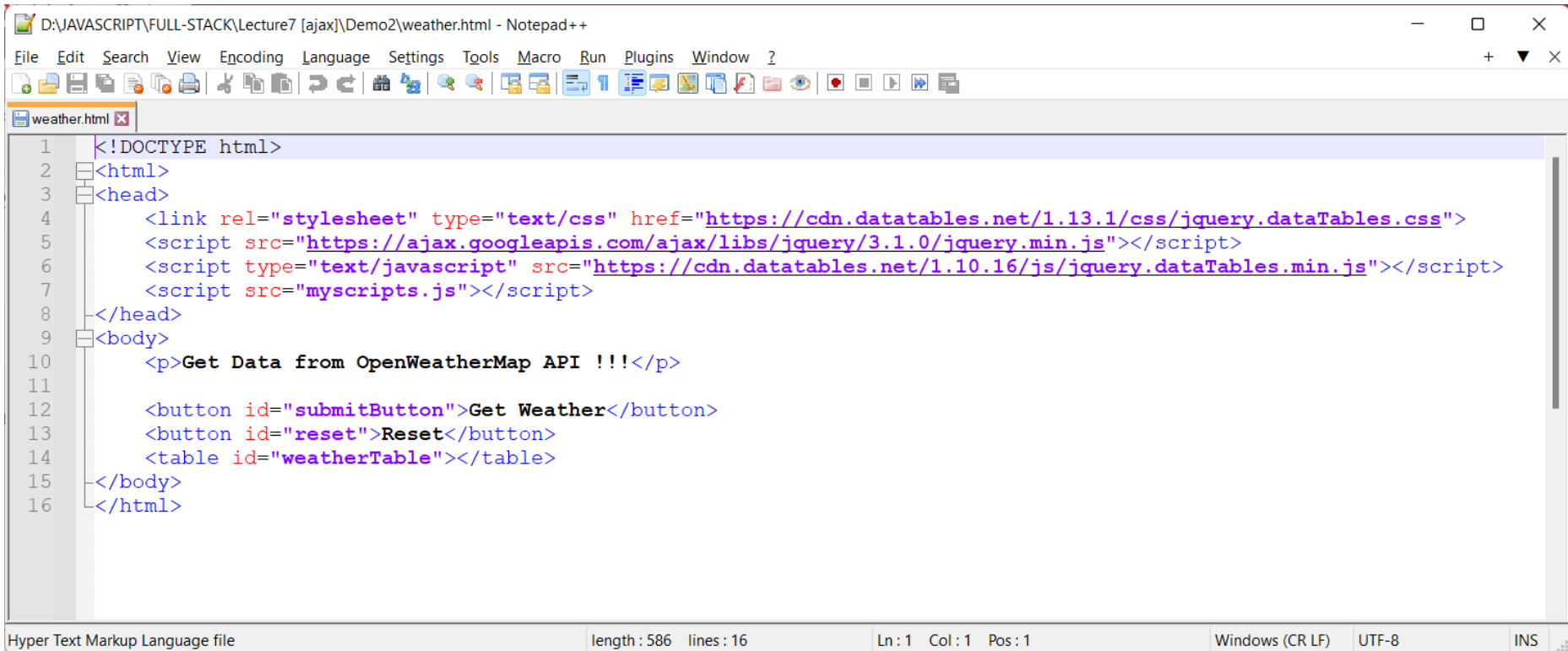
- 1) Open or go to www.openweathermap.org.
- 2) Type the *nearest city* or *town* name into the Search field and click the Search button.
- 3) Locate the city in the search results, and click on its name.
- 4) The City ID can be found in your browser's address bar (*see screenshot below*).
- 5) Example: <https://openweathermap.org/city/1636544>
- 6) Copy this number to the clipboard or write it down.



Working Directory



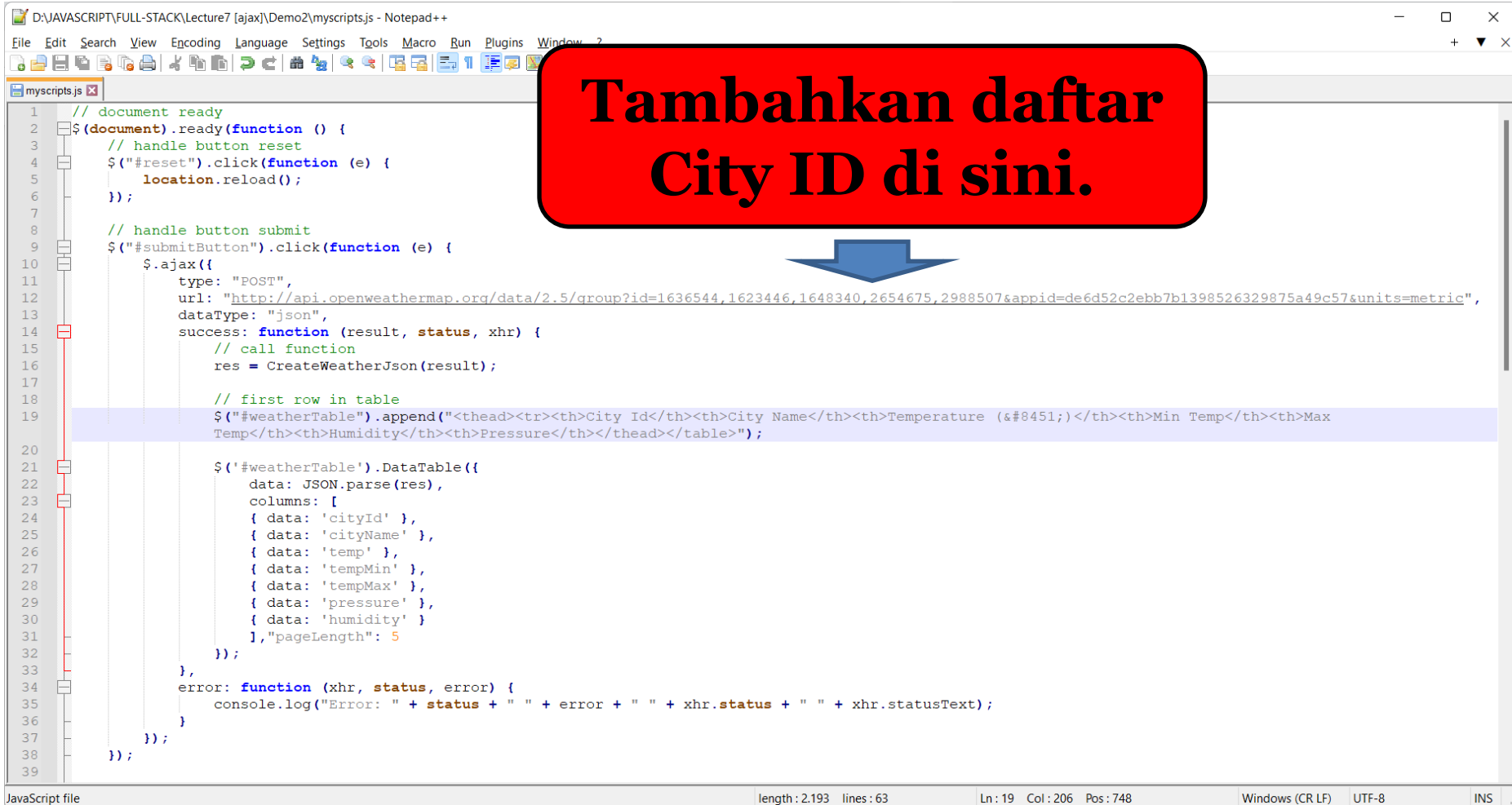
HTML Code



```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <link rel="stylesheet" type="text/css" href="https://cdn.datatables.net/1.13.1/css/jquery.dataTables.css">
5   <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.1.0/jquery.min.js"></script>
6   <script type="text/javascript" src="https://cdn.datatables.net/1.10.16/js/jquery.dataTables.min.js"></script>
7   <script src="myscripts.js"></script>
8 </head>
9 <body>
10   <p>Get Data from OpenWeatherMap API !!!</p>
11
12   <button id="submitButton">Get Weather</button>
13   <button id="reset">Reset</button>
14   <table id="weatherTable"></table>
15 </body>
16 </html>
```

Hyper Text Markup Language file length : 586 lines : 16 Ln : 1 Col : 1 Pos : 1 Windows (CR LF) UTF-8 INS

jQuery (JavaScript Code)



Tambahkan daftar City ID di sini.

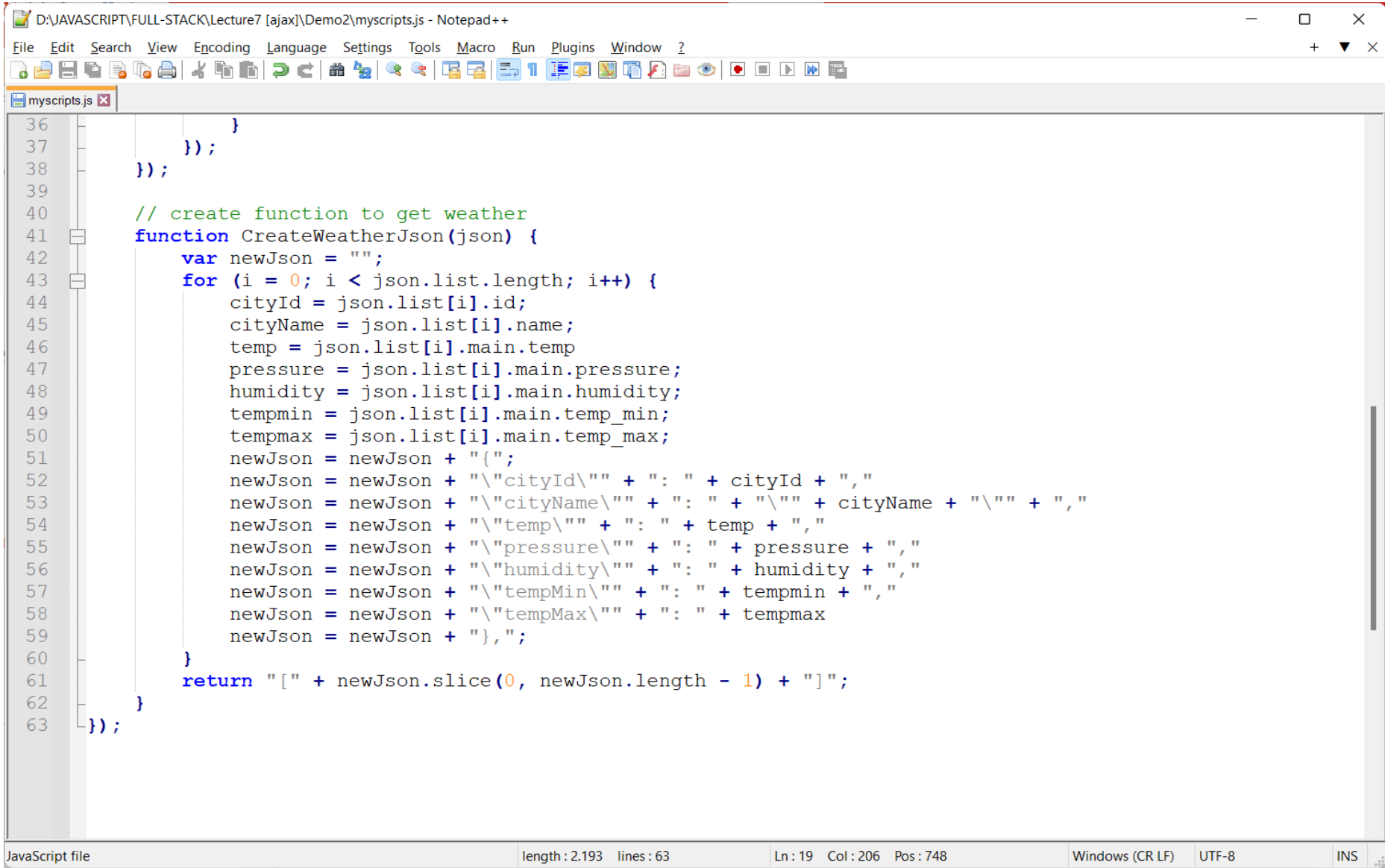
```
1 // document ready
2 $(document).ready(function () {
3     // handle button reset
4     $("#reset").click(function (e) {
5         location.reload();
6     });
7
8     // handle button submit
9     $("#submitButton").click(function (e) {
10        $.ajax({
11            type: "POST",
12            url: "http://api.openweathermap.org/data/2.5/group?id=1636544,1623446,1648340,2654675,2988507&appid=de6d52c2ebb7b1398526329875a49c57&units=metric",
13            dataType: "json",
14            success: function (result, status, xhr) {
15                // call function
16                res = CreateWeatherJson(result);
17
18                // first row in table
19                $("#weatherTable").append("<thead><tr><th>City Id</th><th>City Name</th><th>Temperature (&#451;)</th><th>Min Temp</th><th>Max Temp</th><th>Humidity</th><th>Pressure</th></thead></table>");
20
21                $('#weatherTable').DataTable({
22                    data: JSON.parse(res),
23                    columns: [
24                        { data: 'cityId' },
25                        { data: 'cityName' },
26                        { data: 'temp' },
27                        { data: 'tempMin' },
28                        { data: 'tempMax' },
29                        { data: 'pressure' },
30                        { data: 'humidity' }
31                    ], "pageLength": 5
32                });
33            },
34            error: function (xhr, status, error) {
35                console.log("Error: " + status + " " + error + " " + xhr.status + " " + xhr.statusText);
36            }
37        });
38    });
39 }
```

JavaScript file length : 2.193 lines : 63 Ln : 19 Col : 206 Pos : 748 Windows (CR LF) UTF-8 INS

Openweathermap API:

<http://api.openweathermap.org/data/2.5/group?id=1636544,1623446,1648340,2654675,2988507&appid=d66d52c2ebb7b1398526329875a49c57&units=metric>

jQuery (JavaScript Code)



The image shows a Notepad++ editor window with the title bar "D:\JAVASCRIPT\FULL-STACK\Lecture7 [ajax]\Demo2\myscripts.js - Notepad++". The menu bar includes File, Edit, Search, View, Encoding, Language, Settings, Tools, Macro, Run, Plugins, Window, and Help. The toolbar contains various icons for file operations and editing. The editor is open to a file named "myscripts.js". The code is as follows:

```
36     }
37     });
38 });
39
40 // create function to get weather
41 function CreateWeatherJson(json) {
42     var newJson = "";
43     for (i = 0; i < json.list.length; i++) {
44         cityId = json.list[i].id;
45         cityName = json.list[i].name;
46         temp = json.list[i].main.temp;
47         pressure = json.list[i].main.pressure;
48         humidity = json.list[i].main.humidity;
49         tempmin = json.list[i].main.temp_min;
50         tempmax = json.list[i].main.temp_max;
51         newJson = newJson + "{";
52         newJson = newJson + "\"cityId\"" + ": " + cityId + ",";
53         newJson = newJson + "\"cityName\"" + ": " + "\"" + cityName + "\"" + ",";
54         newJson = newJson + "\"temp\"" + ": " + temp + ",";
55         newJson = newJson + "\"pressure\"" + ": " + pressure + ",";
56         newJson = newJson + "\"humidity\"" + ": " + humidity + ",";
57         newJson = newJson + "\"tempMin\"" + ": " + tempmin + ",";
58         newJson = newJson + "\"tempMax\"" + ": " + tempmax;
59         newJson = newJson + "},";
60     }
61     return "[" + newJson.slice(0, newJson.length - 1) + "]";
62 }
63 });
```

The status bar at the bottom shows "JavaScript file", "length : 2.193", "lines : 63", "Ln : 19", "Col : 206", "Pos : 748", "Windows (CR LF)", "UTF-8", and "INS".

Expected Output

weather.html

File | D:/JAVASCRIPT/FULL-STACK/Lecture7%20[ajax]/Demo2/weather.html

Get Data from OpenWeatherMap API !!!

Get Weather Reset

Show entries Search:

City Id	City Name	Temperature (°C)	Min Temp	Max Temp	Humidity	Pressure
1623446	Tomohon	21.52	21.52	21.52	1010	86
1636544	Manado	26.43	26.43	26.43	1010	94
1648340	Bolaang	25.47	25.47	25.47	1010	88
2654675	Bristol	8.23	7.76	8.9	1027	91
2988507	Paris	4.34	1.81	5.64	1029	90

Showing 1 to 5 of 5 entries

Previous 1 Next

Add More Cities

**Cari dan Tambahkan sekitar 20
City ID ke dalam OpenWeatherMap
API dan tampilkan *output* ke dalam
browser dengan menggunakan AJAX
function di jQuery.**



Exercise #2 *for* Students

REQRES REST-API

URL Real-time API:

<https://reqres.in/>



Test your front-end against a real API

Fake data

No more tedious
sample data creation,
we've got it covered.

Real responses

Develop with real
response codes. GET,
POST, PUT & DELETE
supported.

Always-on

24/7 *free* access in your
development phases.
Go nuts.

HTTP Status Code 404

HTTP Status Code 404 (Not Found)

The screenshot shows the Reqres API interface in a web browser. The browser's address bar displays 'reqres.in'. On the left, a list of API endpoints is visible, each with a 'GET' method button and a description: 'LIST USERS', 'SINGLE USER', 'SINGLE USER NOT FOUND', 'LIST <RESOURCE>', 'SINGLE <RESOURCE>', 'SINGLE <RESOURCE> NOT FOUND', 'CREATE', 'UPDATE', and 'PATCH'. The 'SINGLE <RESOURCE> NOT FOUND' endpoint is currently selected. The main area of the interface is divided into two panels: 'Request' and 'Response'. The 'Request' panel shows the URL '/api/unknown/23' in green text. The 'Response' panel shows the status code '404' in red text, with an empty JSON body '{}'. An advertisement for Namecheap is visible in the bottom left corner, featuring a cartoon character and the text 'Build your website for just \$3.88/mth. More value and performance with Namecheap.' and 'ADS VIA CARBON'.

Reqres - A hosted REST-API read x +

← → ↻ reqres.in ☆ 📄 👤 ⋮

Request
`/api/unknown/23`

Response
404
`{}`

GET LIST USERS

GET SINGLE USER

GET SINGLE USER NOT FOUND

GET LIST <RESOURCE>

GET SINGLE <RESOURCE>

GET SINGLE <RESOURCE> NOT FOUND

CREATE

UPDATE

PATCH UPDATE

Build your website for just \$3.88/mth. More value and performance with Namecheap.

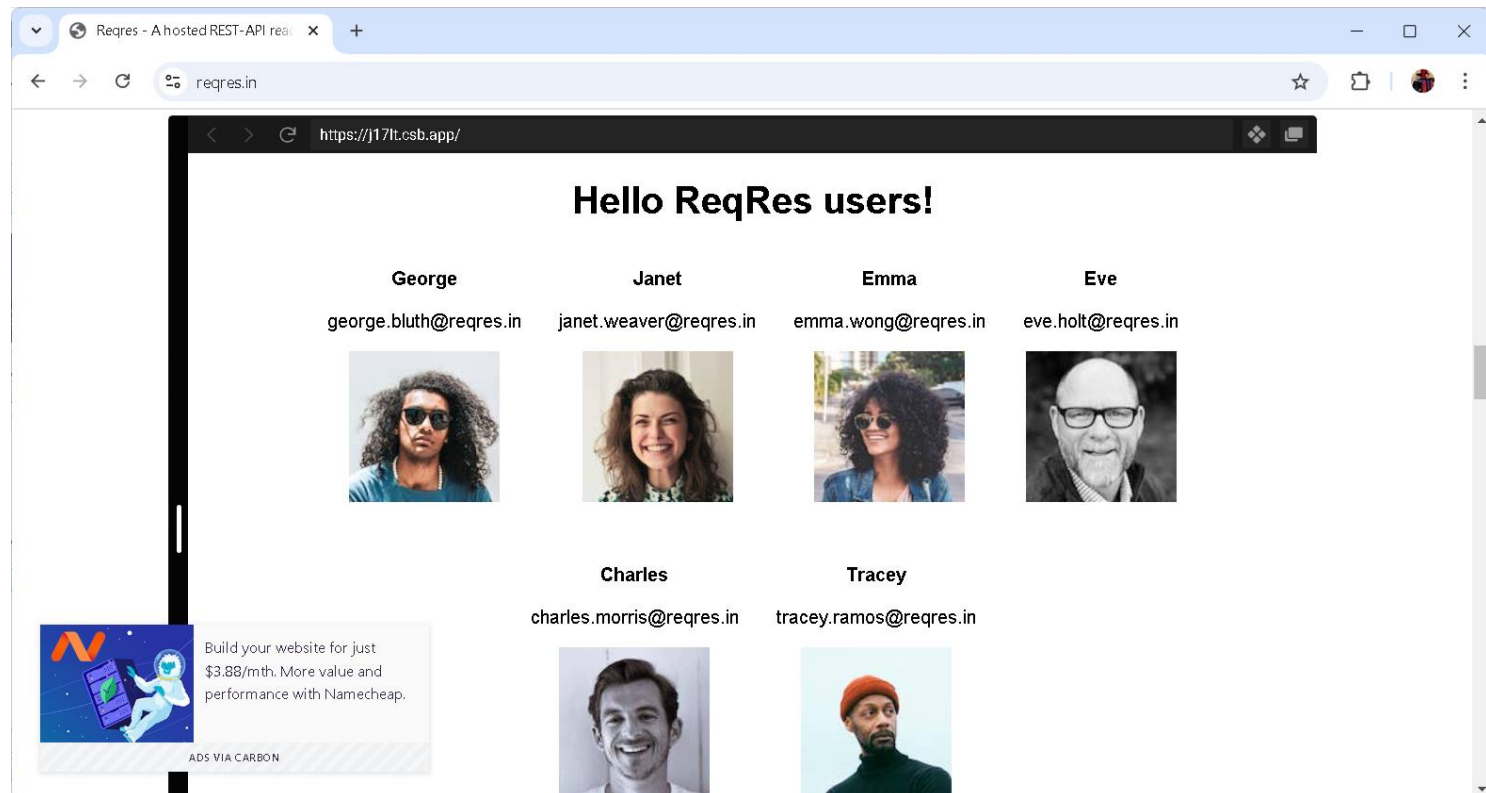
ADS VIA CARBON

API Demo

Gunakan PHP untuk menampilkan data berikut ini di web browser:

URL API:

<https://reqres.in/api/users?page=2>





Exercise #3 *for* Students

HTTP Security Headers

Scan results for https://reqres.in

securityheaders.com/?q=https%3A%2F%2Freqres.in%2F&followRedirects=on

School

Finish update

Home

About

API



Security Headers

by snyk

Scan your site now

Scan

☐ Hide results ☒ Follow redirects

Security Report Summary



Site:	https://reqres.in/
IP Address:	104.26.11.213
Report Time:	28 Jan 2026 22:39:31 UTC
Headers:	✓ Referrer-Policy ✓ X-Content-Type-Options ✓ X-Frame-Options ✗ Strict-Transport-Security ✗ Content-Security-Policy ✗ Permissions-Policy
Advanced:	Not bad... Maybe you should perform a deeper security analysis of your website and APIs: Try Now

HTTP Security Headers

- ❖ HTTP security headers are particularly important for protecting websites and applications. They improve the security of a web server by **preventing or making attacks from outside more difficult**. They can also prevent the disclosure of sensitive data to unauthorized persons. All modern browsers, such as Google Chrome, Mozilla Firefox, Apple Safari, Microsoft Edge, or Opera, support HTTP security headers.
- ❖ HTTP response headers provide directives to the browser's built-in security controls. When we perform a web application **penetration test**, we will examine these headers to determine which browser controls were left open. **This analysis hints at what attack scenarios are more likely** to succeed against the web application.



Security Header adalah aturan keamanan yang dibaca browser, dimana HTTP response header yang dikirim oleh server ke browser untuk:

- ✓ Melindungi website dari serangan umum.
- ✓ Mengatur bagaimana browser berperilaku terhadap konten.
- ✓ Mengurangi risiko XSS, Clickjacking, MIME sniffing, data leakage, dll.

HTTP Security Headers

Prinsip utamanya yaitu security Header akan mengontrol perilaku browser (URL), jadi kita harus mencegah serangan sebelum kode dijalankan (*from url to server*). Ingat juga bahwa Security header bukan sebagai pengganti dari **(1) Secure coding, (2) Validation input, dan (3) Auth yang benar.**

Type	Fungsi	Risiko
Referrer-Policy	Mengontrol informasi URL asal (referrer) yang dikirim browser ke website lain.	URL sensitif (token, ID, query) bisa bocor ke pihak lain.
X-Content-Type-Options	Mencegah browser menebak tipe file (MIME sniffing / Multipurpose Internet Mail Extensions).	XSS, MIME sniffing
X-Frame-Options	Mencegah website ditampilkan di dalam <iframe>.	Clickjacking, User tertipu klik tombol palsu
Strict-Transport-Security (HSTS)	Memaksa browser selalu pakai HTTPS .	Downgrade attack Man-in-the-Middle
Content-Security-Policy (CSP)	Mengatur sumber script, CSS, gambar, iframe, dll.	XSS, Data injection, Script berbahaya
Permissions-Policy / Feature-Policy	Mengatur akses fitur browser modern .	Permissions-Policy: camera=(), microphone=(), geolocation=()

Apa perbedaan **Validation** (*memastikan*), **Sanitization** (*membersihkan*), **Authentication** (*identitas, siapa kamu?*), **Authorization** (*hak akses, bisa buat apa saja?*) di dalam keamanan sistem?

localhost

localhost/xss.php?xss= <script>alert(%27XSS%27)</script>

localhost says
XSS

OK

Connecting...

Console Memory Elements Audits Network Sources Performance Application Security

View: [Icons] Group by frame [] Preserve log [] Disable cache [] Offline Online

Filter [] Hide data URLs All XHR JS CSS Img Media Font Doc WS Manifest Other

Hit Ctrl + R to reload and capture filmstrip.

Name	Headers	Preview	Response	Timing
xss.php?xss=%3Cscript%3Ealert(%27XSS%27)%3C/script%3E	Content-Type: text/html; charset=UTF-8 Date: Fri, 01 Feb 2019 10:04:09 GMT Server: nginx/1.10.3 Transfer-Encoding: chunked X-XSS-Protection: 0			

1 requests | 270 B transferred | Finish: 319 ms

If server sets the **X-XSS-Protection** header to **0** this will disable the XSS Auditor. As a result, we see that the reflected malicious content was rendered.

localhost/xss.php?xss=<script>alert('XSS')</script><h1>This is not blocked because it is not malicious</h1>

Hit Ctrl + R to reload and capture filmstrip.

Network

Filter

Hide data URLs

All

XHR JS CSS Img Media Font Doc WS Manifest Other

1 requests | 317 B transferred | Finish: 8 ms | DOMContentLoaded: 13 ms | Load: 13 ms

Headers

Preview

Response

Timing

Content-Type: text/html; charset=UTF-8

Date: Fri, 01 Feb 2019 10:11:03 GMT

Server: nginx/1.10.3

Transfer-Encoding: chunked

X-XSS-Protection: 1

Request Headers

view source

Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,image/apng,*/*;q=0.8

If server sets the **X-XSS-Protection header to 1**, the document will be loaded but the **malicious content** will just be removed and blocked (*the red part*). Other “non-malicious” content will not be blocked.



Jika
Permissions-Policy
tidak di handle dan
camera=(self),
microphone=(self),
geolocation=(self)
dengan status ON
atau diaktifkan?
Apa yang akan
terjadi?

Tool Audit Keamanan Website

Website untuk audit keamanan website:

- 1) <https://webbkoll.5july.net/en>
- 2) <https://securityheaders.com/>

Pilihlah 10 website per kategori lalu bandingkan (**harus per kategori** contoh *universitas, gov website, hospital, social media, bank, Perusahaan, game online, learning platform, website handphone, dll*), diantaranya **E-Commerce** (Tokopedia, Shopee, Traveloka, IKEA), **Portal Berita** (Detik.com, Kompas.com, Liputan6.com, Kumparan.com, Tribunnews.com), **Search Engine** (Google, Apple, Adobe), Company Indonesia (Pertamina, Nestle, dan Semen Indonesia), dll. Lakukan audit keamanan dari website-website tersebut. Jelaskan temuan dan alasan mengapa beberapa website tidak consider dalam implementasi security header.

1. Berapa persen website yang peduli terhadap keamanan khususnya security header? **Mengapa?**
2. Point apa yang sering menjadi celah keamanan (Referrer-Policy, X-Content-Type-Options, X-Frame-Options, Strict-Transport-Security, Content-Security-Policy atau Permissions-Policy)??? **Mengapa?**
3. Tipe serangan apa yang berpotensi menjadi celah pada website di atas? XSS, Clickjacking, MIME sniffing, data leakage, Man-in-the-Middle (MITM) Attack, Session Hijacking, Cross-Site Data Injection, Unauthorized Browser Feature Access, Downgrade Attack, Phishing & Open Redirect Abuse, Privilege Escalation via Browser API, dll? **Mengapa?**

END PRESENTATION

Thank you for your attention

Instructor: S – W – T

THANK YOU

